

Hamza HOUIR ALAMI

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Engineering student seeking a Summer 2027 Strategy Consulting internship, combining quantitative problem solving, client experience and business development.

Education

Undergraduate Combined GPA: **3.89/4.0**

Arts et Métiers Institute of Technology

Lille / Paris, France

Diplôme d'Ingénieur, MSc-level Grande Ecole Engineering Degree

2025 – Present

- Relevant coursework: Optimization, Probability & Statistics, Numerical Methods, Computer Science, Manufacturing Systems, Industrial Engineering and Applied Mathematics.

Lycée Faidherbe

Lille, France

*Classes Préparatoires aux Grandes Ecoles, PCSI / PSI**

2023 – 2025

- Intensive two-year curriculum in advanced mathematics, physics and engineering science, preparing for nationwide competitive entrance examinations.

Work Experience

Safran Nacelles Morocco

Casablanca, Morocco

Operations & Manufacturing Intern

Summer 2026

- Built a constrained assembly-process optimization model for the Gulfstream G700, structuring operational data, improvement opportunities and process constraints to identify a theoretical cycle-time reduction from **148 to 146 hours**.
- Used Python optimization and a **5,000-scenario Monte Carlo simulation** to evaluate improvement scenarios and quantify their robustness under operational uncertainty.
- Analyzed tooling and production workflows and evaluated process improvements against operational and quality constraints.

Arts et Métiers Junior-Entreprise

Lille, France

Head of Business Development & Partnerships

2026 – Present

- Lead a team of **15 business managers** responsible for client acquisition, project qualification and commercial execution across engineering and consulting engagements.
- Helped expand the commercial pipeline from approximately **600 to 1,600 contacts** and supported the signing of **10 client projects**, representing approximately **€27k** in revenue over three months.
- Translate client needs into project scopes and commercial proposals, coordinating consultants across technical and business engagements.

Mohammed VI Polytechnic University (UM6P)

Rabat, Morocco

AI & Operations Research Intern

Summer 2026

- Developed an end-to-end IT inventory decision-support prototype combining automated data-quality controls, rolling-origin forecasting across nine models and simulation of six inventory policies for 60 department-item series.
- Demonstrated, on synthetic validation data, a potential **21.2% reduction in total costs** and **92.5% fewer emergency units** compared with a reactive purchasing policy; designed the pipeline for recalibration on real UM6P operational data.

Selected Projects

Decathlon — Operations & Data Transformation

Consultant, AMJE Lille

- Translated client requirements into redesigned data structures and reconciliation rules for a global product-data tool, improving traceability, duplicate control and usability.

ThePnLab — Algorithmic Paper-Trading Platform

Python, FastAPI, PostgreSQL

- Built and deployed an educational paper-trading platform combining 30-minute market screening, multi-signal decision support, position sizing, portfolio tracking and market-risk regime detection.

CNN-Based French Speech Recognition

Python, CNN, Signal Processing

- Built a lightweight CNN using Mel spectrograms, achieving **95.2% accuracy** on the initial single-token classification task; extended the system to spoken French numbers from 0 to 99 using silence-based segmentation and grammar-constrained decoding.

Flywheel Geometry & Energy-System Optimization

Python, Applied Mathematics

- Proved the circular fixed-perimeter geometry as the global optimum for rotational inertia using Wirtinger's inequality, then implemented a reproducible Python model and audited Tracker experiments to identify an initial-speed confound.

Leadership, Skills and Interests

- **Award:** Bourse Genin 2026 laureate, an excellence and integration scholarship for international students in France.
- **Leadership:** Mentoring Moroccan engineering students; former competitive football player (Ibercup Cascais 2017).
- **Analytics & Methods:** Excel, Python, quantitative analysis, optimization, Monte Carlo simulation, forecasting, process analysis and structured problem solving.
- **Engineering Tools:** CATIA / 3DEXPERIENCE, Abaqus / FEA, STAR-CCM+ and SolidWorks.
- **Languages:** Arabic (native), French (fluent) and English (advanced).
- **Interests:** Financial markets, football, Formula 1, geopolitics and travel.